

EX. NO: 2 — PLAYFAIR CIPHER

Aim:

To implement the Playfair Cipher substitution technique using C.

Algorithm:

1. Read plaintext and key.
2. Construct a 5×5 key matrix.
3. Combine I and J.
4. Divide plaintext into pairs.
5. Apply Playfair encryption rules.
6. Display ciphertext.

Program:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
char keymat[5][5];
void createMatrix(char key[])
{
    int used[26] = {0};
    int i, j, k = 0;
    char ch;
    used['J' - 'A'] = 1;
    for(i = 0; key[i] != '\0'; i++)
    {
        ch = toupper(key[i]);
        if(ch == 'J')
            ch = 'I';
        if(ch >= 'A' && ch <= 'Z' && !used[ch-'A'])
        {
```

```

        keymat[k/5][k%5] = ch;
        used[ch-'A'] = 1;
        k++;
    }
}
for(ch = 'A'; ch <= 'Z'; ch++)
{
    if(!used[ch-'A'])
    {
        keymat[k/5][k%5] = ch;
        k++;
    }
}
}
void find(char ch, int *r, int *c)
{
    int i, j;
    if(ch == 'J')
        ch = 'I';
    for(i = 0; i < 5; i++)
        for(j = 0; j < 5; j++)
            if(keymat[i][j] == ch)
            {
                *r = i;
                *c = j;
                return;
            }
}
}

```

```

int main()
{
    char key[50], text[100], a, b;
    int i, r1, c1, r2, c2;
    printf("Enter key: ");
    scanf("%s", key);
    printf("Enter plain text: ");
    scanf("%s", text);
    createMatrix(key);
    printf("\nKey Matrix:\n");
    for(i = 0; i < 5; i++)
    {
        for(int j = 0; j < 5; j++)
            printf("%c ", keymat[i][j]);
        printf("\n");
    }
    printf("\nCipher text: ");
    for(i = 0; text[i] != '\0'; i += 2)
    {
        a = toupper(text[i]);
        b = text[i+1] ? toupper(text[i+1]) : 'X';
        if(a == 'J') a = 'I';
        if(b == 'J') b = 'I';
        if(a == b)
        {
            b = 'X';
            i--;
        }
    }
}

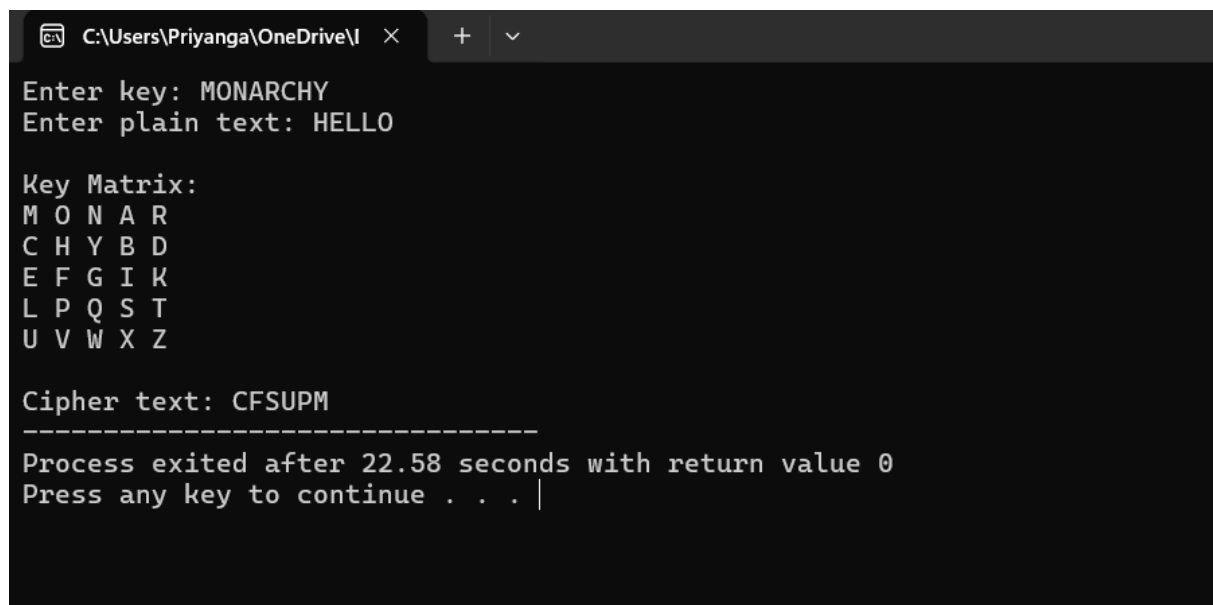
```

```

    find(a, &r1, &c1);
    find(b, &r2, &c2);
    if(r1 == r2)
        printf("%c%c", keymat[r1][(c1+1)%5],
                keymat[r2][(c2+1)%5]);
    else if(c1 == c2)
        printf("%c%c", keymat[(r1+1)%5][c1],
                keymat[(r2+1)%5][c2]);
    else
        printf("%c%c", keymat[r1][c2],
                keymat[r2][c1]);
}
return 0;
}

```

Output:



```

C:\Users\Priyanga\OneDrive\I >
Enter key: MONARCHY
Enter plain text: HELLO

Key Matrix:
M O N A R
C H Y B D
E F G I K
L P Q S T
U V W X Z

Cipher text: CFSUPM
-----
Process exited after 22.58 seconds with return value 0
Press any key to continue . . .

```

Result:

Thus, the Playfair Cipher substitution technique was implemented successfully using C.